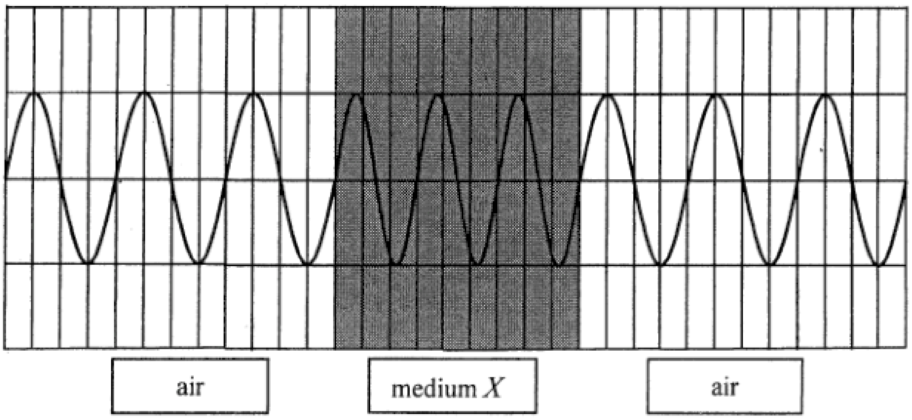
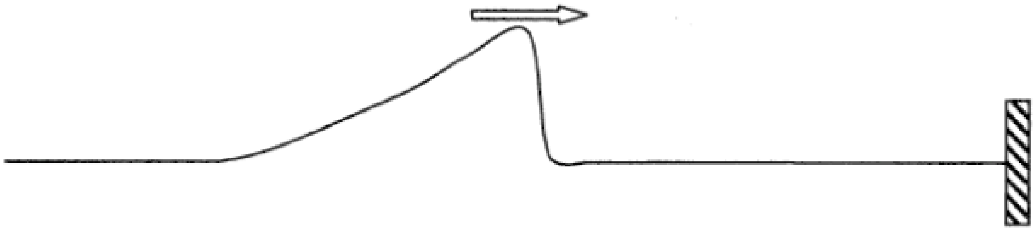
2012 Q17

17. A certain monochromatic light passes through medium $X$ as shown below. What is the refractive index of medium $X$ ?



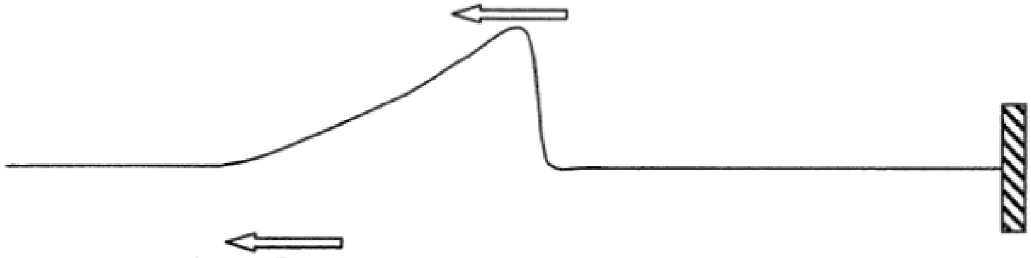

A. 1.25
B. 1.33
C. 1.50
D. 1.65

16. A pulse on a string propagates towards the right end which is fixed.

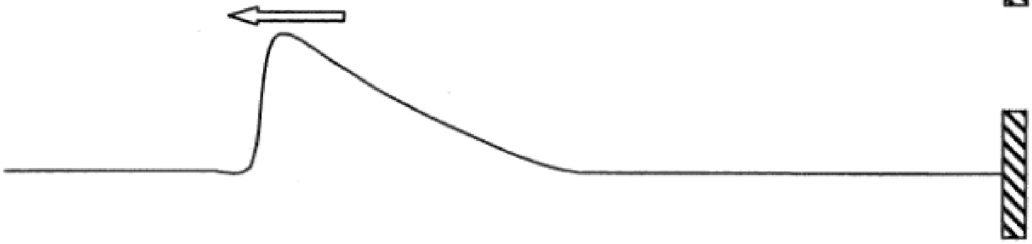


Which of the following represents the reflected pulse ?

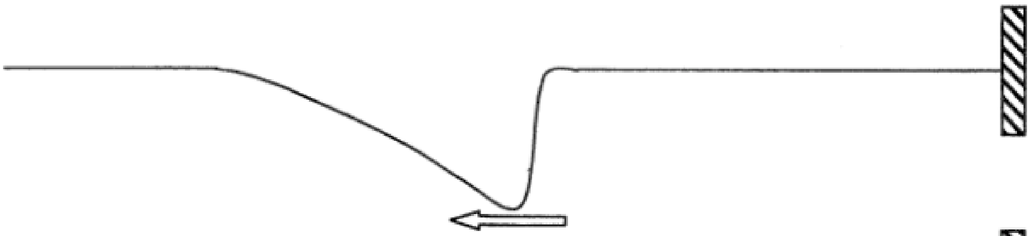
A.



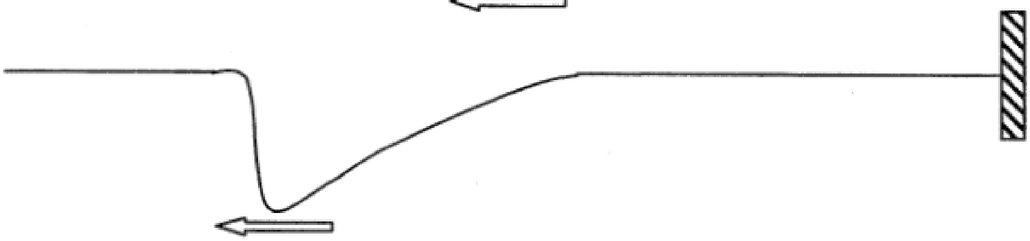
B.



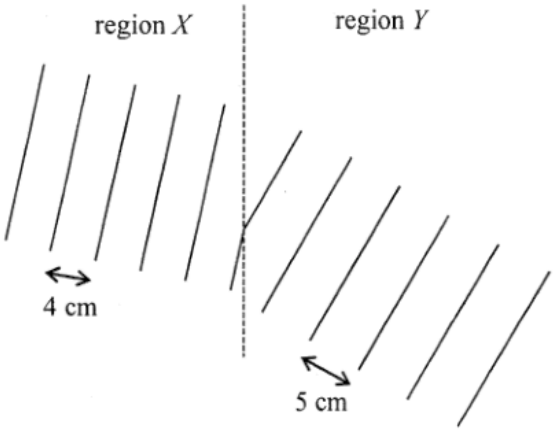
C.



D.



16. The figure shows plane water waves travelling from region $X$ to region $Y$. The wavelengths of the water waves in regions $X$ and $Y$ are 4 cm and 5 cm respectively.

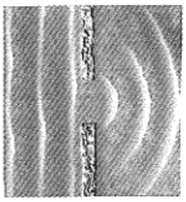



Which of the following statements is correct ?

A. The speed of the water waves in region $X$ is higher than that in region $Y$.
B. The direction of travel of the water waves bends towards the normal as they enter region $Y$.
C. The frequency of the water waves is the same in both regions.
D. If plane water waves of wavelength 5 cm travel from region $Y$ to region $X$, the wavelength becomes 6 cm after the waves enter region $X$.



16.

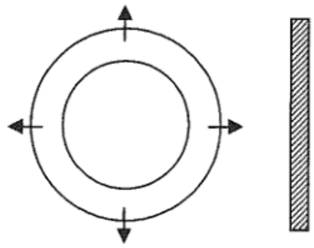


The photograph shows a series of plane sea waves travelling through a gap in a sea wall which exhibits diffraction. Assuming that the frequency of the waves remains unchanged, which of the following will increase the degree of diffraction ?

(1) The gap in the sea wall becomes narrower.
(2) The wavelength of the waves increases.
(3) The amplitude of the waves becomes larger.

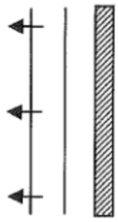
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

13.

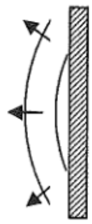


The above figure shows two circular pulses produced by drops of water falling on a ripple tank. The pulses are then reflected by a straight barrier. Which diagram best shows the reflected pulses ?

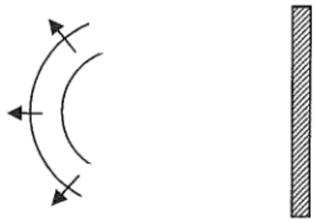
A.



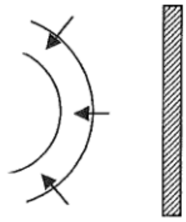
B.



C.



D.

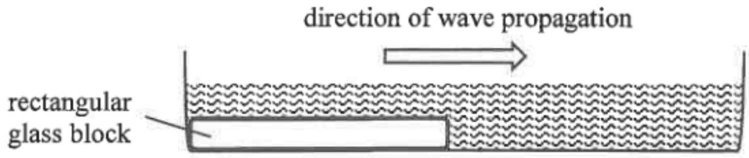


2020 Q14

*14. A light beam consisting of wavelengths $\lambda_1$ and $\lambda_2$ is incident normally on a diffraction grating. The third-order diffraction of wavelength $\lambda_1$ coincides with the fourth-order diffraction of wavelength $\lambda_2$ in the resulting pattern. If $\lambda_1$ is 680 nm, find $\lambda_2$.

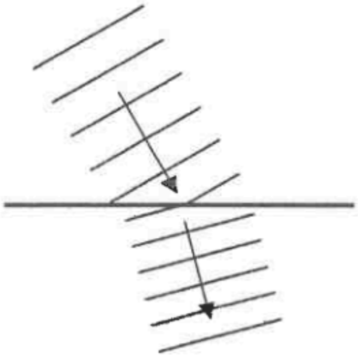
A. 510 nm
B. 680 nm
C. 907 nm
D. It cannot be determined because the grating spacing is unknown.

17. Plane water waves travel from shallow water to deep water in a ripple tank as shown below.

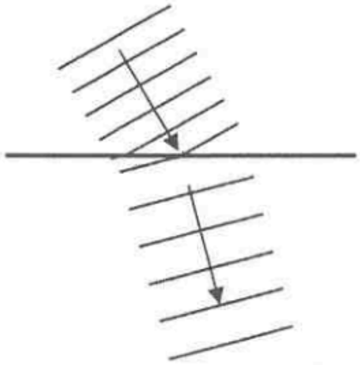



Which diagram correctly shows the top view of the wavefronts in the ripple tank ?

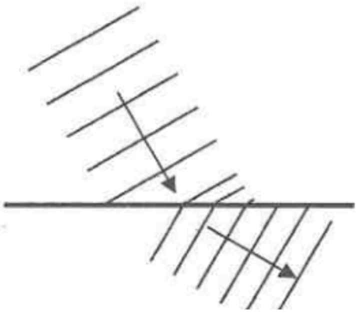
A.



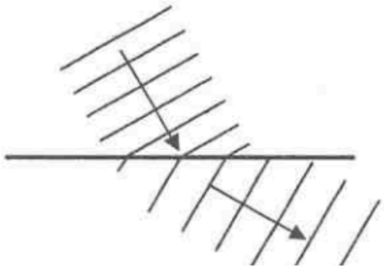
B.



C.



D.



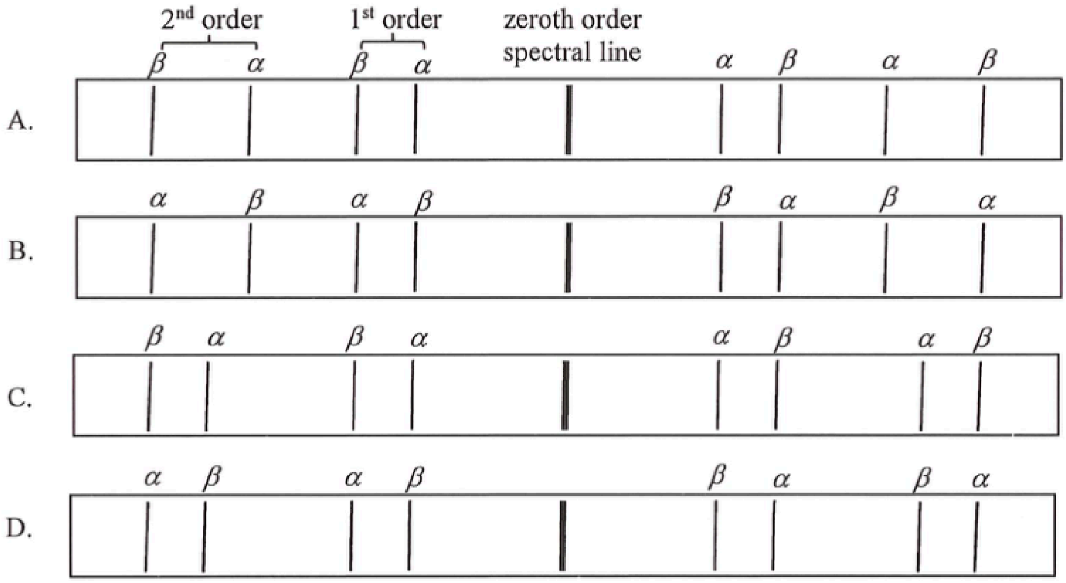


*20. For a diffraction grating of 600 lines per mm, the diffracted red light (657 nm) coincides with the diffracted violet light (438 nm) at angle of diffraction 52°. What are the respective orders of the diffracted red light and violet light ?

| | **red** | **violet** |
|---|---|---|
| A. | 2 | 3 |
| B. | 3 | 4 |
| C. | 3 | 2 |
| D. | 4 | 3 |

2023 Q18

18. A diffraction grating is used to observe the $\alpha$ (656 nm) and $\beta$ (486 nm) lines emitted from a gas discharge tube. Which of the following best represents the pattern observed ?



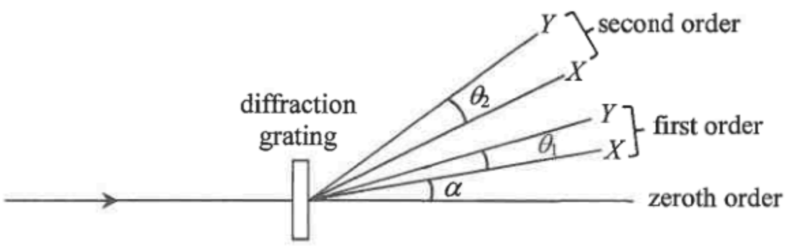

2025 Q17

*17. The light from a sodium lamp consists of two wavelengths 589.00 nm and 589.59 nm. Find the angle between these two spectral lines in the second-order after the light passes through a diffraction grating of 600 lines per mm.

A. 0.058°
B. 0.041°
C. 0.022°
D. 0.021°

2022 Q18

*18.




A light beam which consists of monochromatic lights $X$ and $Y$ is incident normally on a diffraction grating. The figure shows the spectra of the first two orders. The diffraction angle of $X$ in the first order is $\alpha$. The angular separation between $X$ and $Y$ in the first order is $\theta_1$ and that in the second order is $\theta_2$. Which statement **must be** correct ?

A. The wavelength of $Y$ is shorter than that of $X$.
B. $\alpha$ would be larger if the grating spacing is smaller.
C. $\theta_1$ is independent of the grating spacing.
D. $\theta_2 = 2\theta_1$

2016 Q19

19. Diffraction will occur when light

(1) passes through a pinhole.
(2) passes by a sharp edge.
(3) passes through a slit.

A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

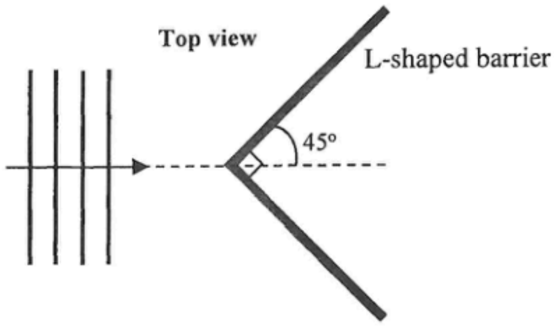
2025 Q21

21. Which of the following phenomena can be explained by the speed of waves being different in different media ?

(1) echoes
(2) rainbows
(3) mirages

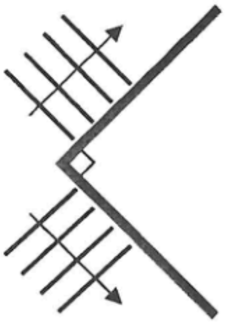
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

15.

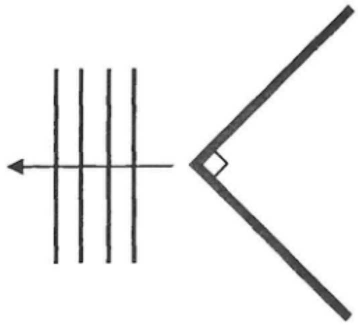



In a ripple tank, a series of plane water waves approach an L-shaped barrier as shown above. Which diagram below best shows the wave pattern reflected from the barrier ?

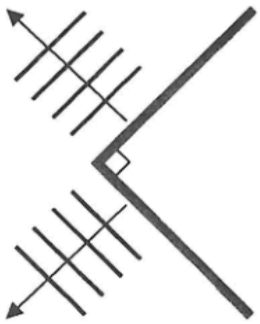
A.



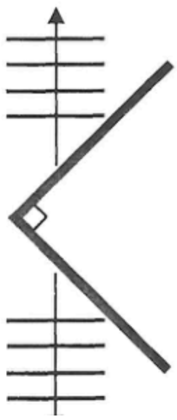
B.



C.

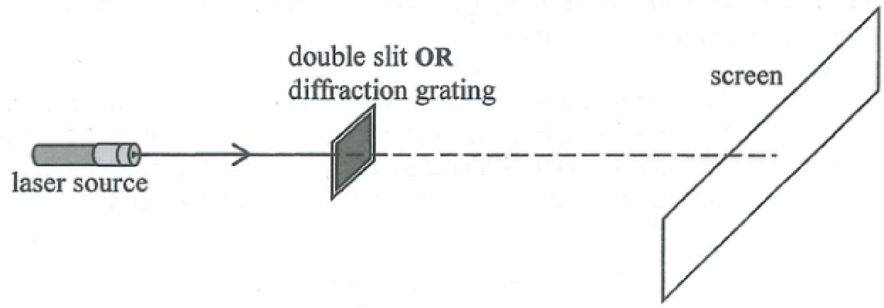


D.

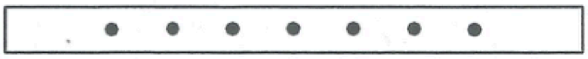


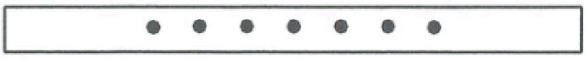
2019 Q19

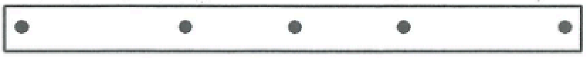
19.

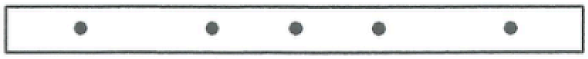



A double slit and a diffraction grating are used in turns in the above set-up such that red and green laser lights are directed one after the other on each of them. The four resulting patterns of bright spots obtained on the screen are shown below. Which one belongs to the pattern formed by **green light** incident on the **diffraction grating** ?

A. 

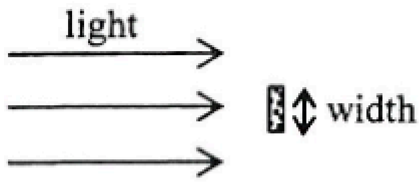
B. 

C. 

D. 

2018 Q16

16. Light undergoes diffraction round an obstacle.



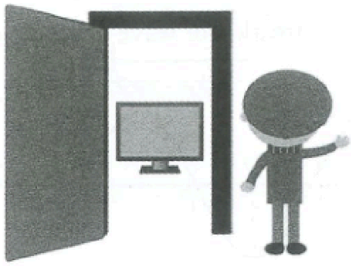

The angle of diffraction would increase when

(1) the amplitude of the incident light is increased.
(2) the width of the obstacle is increased.
(3) the wavelength of the incident light is increased.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2019 Q17

17. Peter is standing next to the door of a room. He can hear the sound emitted by the television inside the room but cannot see the television pictures. Which of the following is/are the possible reason(s) ?

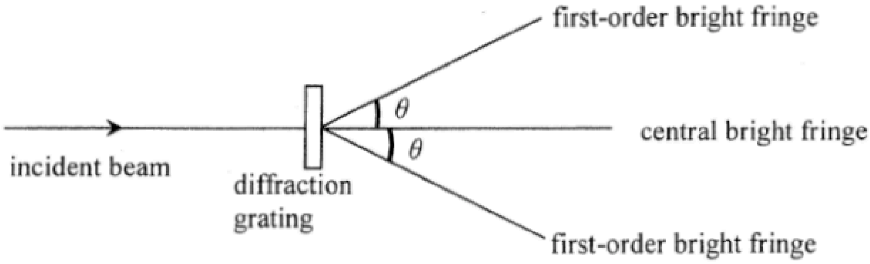


(1) Sound wave diffracts while light wave does not.
(2) Sound wave is mechanical in nature while light wave is electromagnetic.
(3) Sound wave has a much longer wavelength than visible light.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2013 Q23

*23.

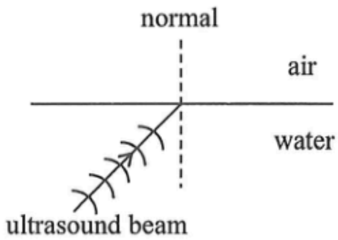



When monochromatic light passes through a diffraction grating, a pattern of bright fringes is formed. Which arrangement would produce the greatest angle $\theta$ between the central and first-order bright fringes ?

| | grating (lines per mm) | colour of light |
|---|---|---|
| A. | 400 | green |
| B. | 400 | blue |
| C. | 200 | green |
| D. | 200 | blue |

2025 Q20

20. A source in water emits a beam of ultrasound towards the water-air interface as shown.

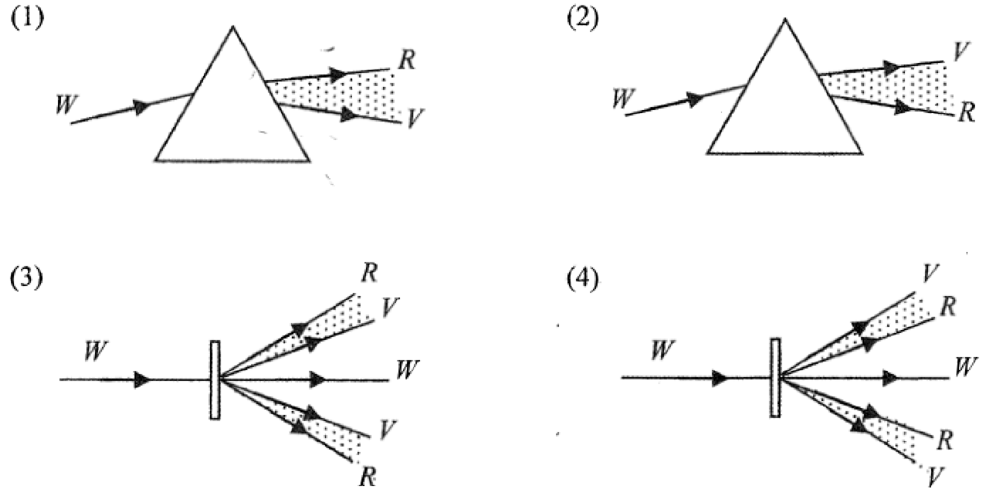



Which of the following statements is **INCORRECT** ?

A. The frequency of the refracted beam is the same as that of the incident beam.
B. The speed of the refracted beam becomes smaller.
C. The refracted beam leaving the interface bends towards the normal.
D. If the angle of incidence is large enough, total internal reflection may occur.

2015 Q17

17. Which diagrams below correctly show the spectra formed from white light by a glass prism and a diffraction grating respectively ? It is known that red light travels faster than violet light in glass. ($R$ = red, $V$ = violet, $W$ = white)




A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

2017 Q17

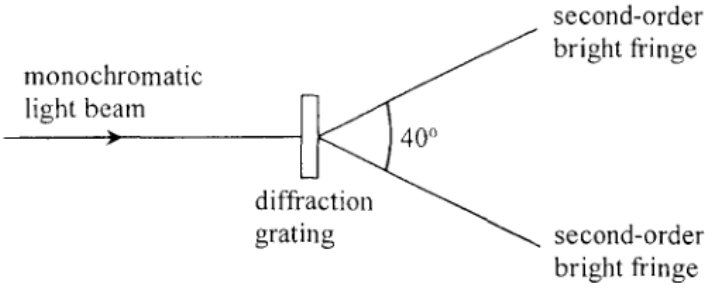
17. In which of the following situations **MUST** the direction of travel of a wave change ?

(1) when a wave is reflected by a barrier
(2) when a wave enters from one medium to another medium
(3) when a wave travels through a gap smaller than its wavelength

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2021 Q17

*17.




When a monochromatic light beam is incident normally on a diffraction grating with 300 lines per mm, a pattern of bright fringes is formed. The angle between the two second-order bright fringes is 40°. Find the frequency of the light.

A. $1.4 \times 10^{14}$ Hz
B. $2.6 \times 10^{14}$ Hz
C. $2.8 \times 10^{14}$ Hz
D. $5.3 \times 10^{14}$ Hz

2025 Q18

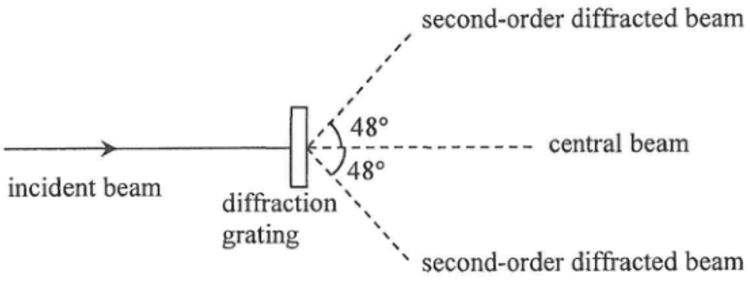
18. In an experiment, monochromatic light is incident normally on a double slit and a diffraction grating placed in turn at the same position. The respective interference and diffraction patterns formed on the screen are observed. Which of the following statements about the patterns is/are correct ?

(1) The fringe separation of the interference pattern is wider.
(2) The fringes of the diffraction pattern are sharper.
(3) The width of the whole diffraction pattern is wider.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2026 Q20

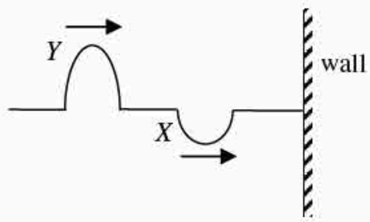
*20. A beam of monochromatic light is incident normally on a diffraction grating. Second-order diffracted beams are found at 48° to the incident direction as shown in the figure. Find the highest order of diffracted beam that can be observed.




A. 2nd order
B. 3rd order
C. 4th order
D. 5th order

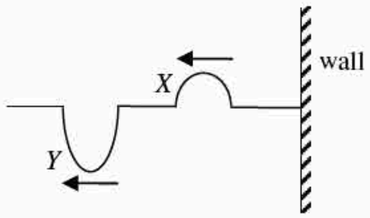
pp Q18

18.

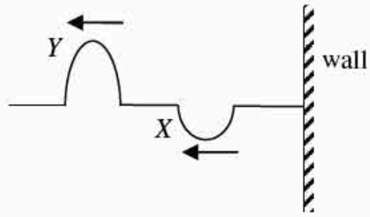



Two pulses, $X$ and $Y$, are travelling along a string which is fixed at one end to the wall as shown in the figure above. Which of the following is a possible waveform of the string after the two pulses reflect ?

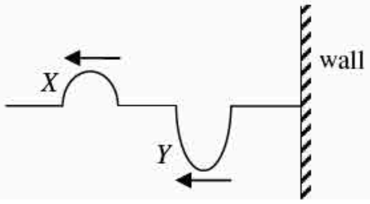
A.



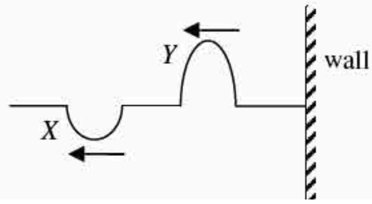

B.




C.




D.



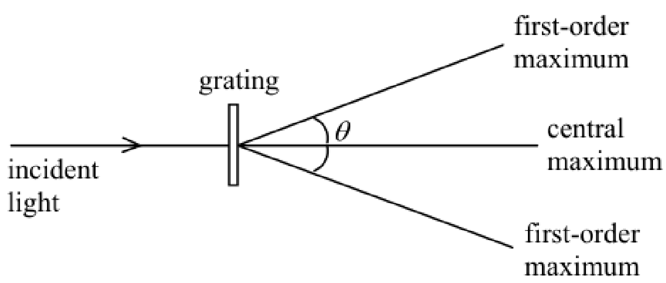

pp Q23

*23. Yellow light of wavelength 590 nm is incident normally on a diffraction grating with 400 lines per mm. Find the difference in angular positions for the third order and the fourth order bright fringes.

A. 13.7°
B. 25.7°
C. 45.1°
D. 70.7°

sap Q23

23.




When monochromatic light is passed through a diffraction grating, a pattern of maxima and minima is observed as shown. Which combination would produce the largest angle $\theta$ between the first-order maxima ?

| | **Grating (lines per mm)** | **Colour of light used** |
|---|---|---|
| A. | 200 | blue |
| B. | 200 | red |
| C. | 400 | blue |
| D. | 400 | red |